Zero Shot Sentiment Analysis on Tweets in Any Language

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Objective

We train a sentiment analysis convolutional neural network classification model on a set of sentiment-labeled tweets in a popular language, and aim to transfer learn to classify sentiment in any other language given a set of unlabeled tweets in that language.

Introduction

In order to meet our objective above, we re-purpose the Unicode Convolutional Neural Network [1] which offers two advantages.

1. It pre-trained to solve the geolocation problem, which has massive amounts of reliable data.
2. It is already capable of analyzing text in any language by looking at its underlying representation in Unicode.



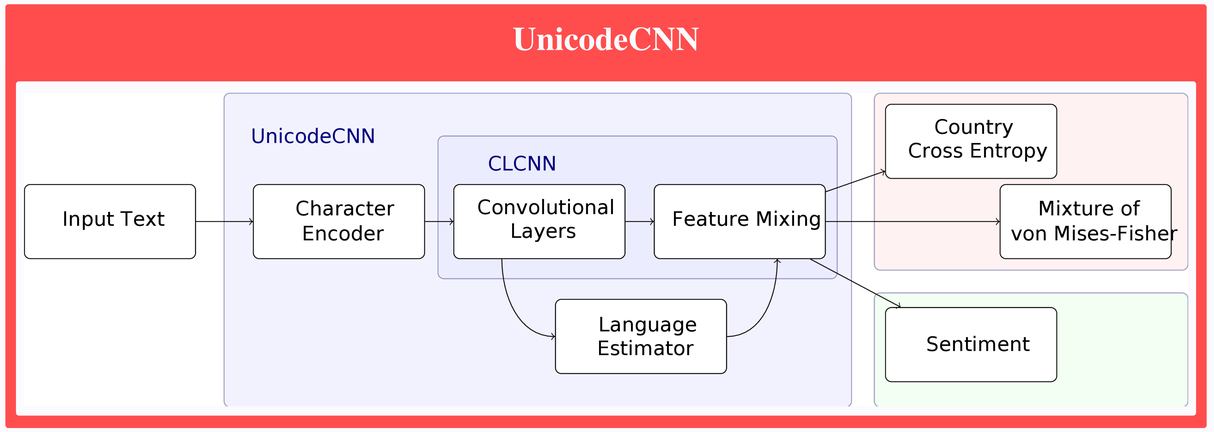
(a) Portuguese tweet whose sentiment we'd like to evaluate



(b) Similar English tweet with known sentiment

Figure: Tweets Similar in Structure and Sentiment

UnicodeCNN



Dataset

The majority of the natural language processing community has focused on analyzing English text since it comprises roughly 42% of all activity on Twitter, which is unfortunate since there are over 100 languages actively used on the platform spread over the world as shown below.

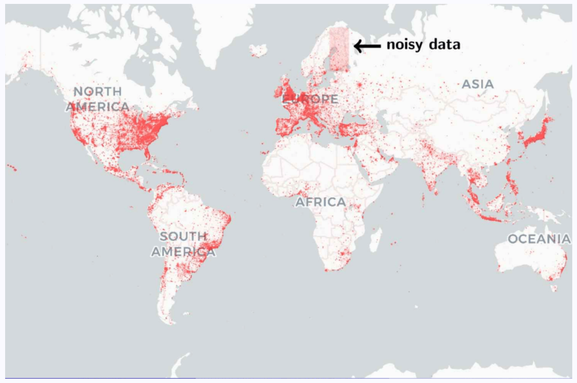


Figure: Global Twitter Activity

Since it is impractical to generate significant amounts of sentiment-labeled data for all languages, we use a publicly available sentiment-labeled dataset [2] containing 1.6 million tweets in 13 languages. While the data labels are precise for some *high resource languages* like English, the figure on the right shows that the data is unreliable for most other languages.

Model

In this experiment, we will use the UnicodeCNN pretrained to solve the geolocation problem on all 900 million geo-encoded tweets sent over 9 months by over 3.0 million users in over 100 languages, and replace the final MvMF estimation layer with a layer of sentiment classification as shown above. We will retrain some layers of the model on a training set and evaluate on a testing set, where the languages of the two sets may or may not agree.

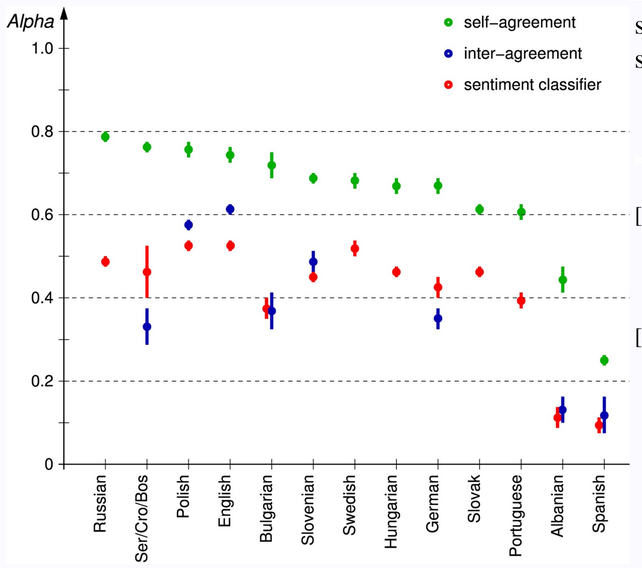


Figure: Label agreement across annotators

Contribution

In the main experiments highlighted in the table below, we train on sentiment-labeled English tweets, test on unlabeled Portuguese tweets, and validate against a model trained on sentiment-labeled Portuguese tweets.

#	Retrain	Train	Test
1	All	English	English
2	All	Portuguese	Portuguese
3	All	English	Portuguese
4	Final	English	English
5	Final	Portuguese	Portuguese
6	Final	English	Portuguese

Figure: Main experiments highlighted.

We chose English and Portuguese so that we could train the main model and the validation model with high-quality data, but our tests did not require the Portuguese sentiment-labels to classify sentiment on Portuguese tweets.

After finalizing the results on Portuguese, we would like to test and validate this model on tweets in other languages. If successful, we would expect the English-trained model to succeed on other low resource languages for which significant amounts of sentiment-labeled data do not exist.

References

[1] Mike Izbicki, Evangelos Papalexakis, and Vassilis Tsotras.
Geolocating tweets in any language at any location.
CIKM, 2019.

[2] Igor Mozetič, Miha Grčar, and Jasmina Smailović.
Multilingual twitter sentiment classification: The role of human annotators.
PLOS ONE, 11(5):1–26, 05 2016.

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